

Q.	Solution
1(a)	1. The resultant forces acting on the system must be zero. 2. The resultant torque about any axis must be zero.
1(b)(i)	Yes. Since the ball is falling at a constant speed vertically downwards, there is no change in velocity, hence there is no acceleration.
(ii)	No. The resultant force acting on the satellite is the gravitational force. This results in the satellite accelerating continuously towards the earth, as the direction of the velocity is constantly changing.
2(a)	$x = \frac{\lambda D}{\sin \theta} \rightarrow (1.4 \times 10^{-3}) = \frac{\lambda(2.1)}{\sin 30^\circ}$